- A set of sampled data points from each of the receive channels
  Each data point is complex data. (The sampled data itself is quadrature encoded, decoded by the receive chain hardware into I and Q value pairs.)
- Header values that describe the hardware status at the time of acquisition (transmit gain, center frequency, analog receive gain, and hardware fault status)
- Header with the peak sampled data for each receive channel, as seen by the hardware
  This is commonly called the A-to-D peak.

PSS sends control data to MR Recon, passed through the receive chain. These are called DAB packets (a legacy name from "Data Acquisition Board" packets, even though the Data Acquisition Board has not been in the Scanner for years). Most DAB packets tell MR Recon how to interpret the acquired raw data frames by specifying their location in k-space. MR Recon uses this information to know how to organize (sort) incoming frame data into MR k-space matrices that can later be processed. A PSD may play the same excitation multiple times and command MR Recon to sum such frames into the k-space matrices rather than just copy them. This is called NEX'ing (NEX comes from Number of EXcitations).

There are also DAB packets that are strictly for control (pass done, scan done, region done, etc.), used to initiate actions in MR Recon.

Faults in the receive chain hardware are sent to MR Recon, and MR Recon aborts the scan if hardware faults are indicated.

The following table summarizes the communications between the subsystems.

| From | To | Data | Interface Standard |
|---|---|---|---|
| Receive Chain | MR Recon | control data packets | comm. link and DMA |
| Receive Chain | MR Recon | raw data frames | comm. link and DMA |
| MR Recon | Receive Chain | setup and control data | programmed IO and comm. link |

### 5.4.9. MR Recon & SCP [I]

MR Recon receives a Scan Prep event prior to starting the scan. MR Recon is expected to allocate acquisition memory and other any other resources that are needed to complete the scan. MR Recon sends a success/failure message back scan control at the completion of scan prep.

MR Recon is also involved in stopping the scan, either a normal stop or an abort. Normal stops are sent from MR Recon to scan control at the end of each Prescan pass and also at the end of a normal scan.

MR Recon can initiate a scan abort when the receive chain faults or due to an internal reconstruction fault. An MR Recon-initiated scan abort sends an abort command to SCP.

When SCP detects a scan abort for any reason (scanner hardware fault, operator manually stopped the scan, MR Recon-initiated abort), it communicates an abort stop command to MR



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Recon. MR Recon shuts down the receive chain and responds to scan control when that shutdown finishes.

### 5.4.10. MR Recon & PSI [J]

Real-time Sequence Parameters (RSPs) from PSS to MR Recon are used for real-time scans. An RSP packet contains prescription parameters that may change during the scan. MR Recon reads RSPs throughout a real-time scan to control the processing steps to be performed on the acquired data.

There is an "Auto-Pause" control interface between MR Recon and PSS that allows MR Recon to slow the PSD play out if MR Recon is unable to keep up. This is used in scans in which MR Recon does not pre-allocate all of the memory to hold incoming acquired data frames.

MR Recon includes the capability to analyze the acquired data and provide feedback to PSS for controlling PSDs. This capability is called Real-Time Processing, or RTP. Control data from the PSD is sent to the RTP server in MR Recon, which is then available to the RTP Application. The RTP feedback uses a low latency data path from MR Recon to PSS, allowing PSS to apply changes in the pulse and sequence logic of PSDs as the scan is running. Examples of Real-Time Processing Applications using this interface include Navigator, Echo Train Alignment, and Prospective Motion Control (PROMO).

### 5.4.11. MR Recon & PSS [K]

PSS sends control/configuration data to MR Recon for specific applications. This information supplements information that is normally sent in DAB packets but is too complex to fit in the standard DAB packet structures.

### 5.4.12. Host & MR Recon [L]

Host collects prescription data from the user interface, MR Recon's Prescription Server, and the Host PSD. It manipulates the data into the form needed by MR Recon and stores the information on disk. This information is subsequently transferred to MR Recon via socket I/O.

MR Recon also acts as the broker in communicating scanner state between Host and SCP. It forwards scan start events from Host to the MR Scanning Hardware. Control events from the SCIM (start/stop/pause) are communicated through MR Recon to Host.

MR Recon prepares for each scan by downloading scan specific parameters from Host. This includes Prescan setup structures and Image Reconstruction setup.

The Host prescan process interacts with MR Recon's prescan. For Auto-Prescan, Host initiates Prescan and expects a pass/fail response. For Manual Prescan, Host displays a user interface, and the user enters/confirms values that are forwarded to the MR Recon prescan in real-time to affect the Prescan operation and set values on the MR Scanning Hardware. The MR Recon prescan continually processes acquired frames and sends results to the Host prescan for display on the user interface.



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In the course of scanning a patient, MR Recon sends image data to Host. MR Recon image data is composed of the raw pixel values, a fixed header, and an optional extensible segment for application specific parameters.

The following table summarizes of the communications between the subsystems.

| From | To | Data | Interface Standard |
|---|---|---|---|
| Host | MR Recon | prescription controls | TCP/IP & MrMail |
| Host | MR Recon | prescan controls | TCP/IP & MrMail |
| MR Recon | Host | prescan parameters<br>scan parameters | TCP/IP & MrMail |
| MR Recon | Host | image data | MrMail & Memory server |

### 5.4.13. Host & Host Platform [M]

During startup, Host Platform initiates Host components and communicates startup state information to be used in managing an ordered startup of the Scanner. Host will log all system messages through the Host Platform message interface.

Host Platform communicates session operations & task operations to Host. It also communicates patient information, user preferences and protocol. It is also responsible for managing the Prescription and Acquisition User Interface owned by Host. It receives images generated by Host for storage and further processing.

The following table summarizes the communications between the subsystems.

| From | To | Data | Interface Standard |
|---|---|---|---|
| Host | Host Platform | DICOM images | File I/O and messages |
| Host Platform | Host | patient information, preferences, session/task operations | GEOCOMM |
| Host | Host Platform | system, startup, and status messages | TCP/IP & MrMail |

### 5.4.14. Host Platform & Viz [N]

This section expands on the interfaces surrounding Viz. Though there are visualization components in other phases of workflow, such as graphic prescription, this section focuses primarily on the post-processing phase.

Invalid checksums associated with the image data consistency checks are considered unrecoverable errors; no processing or creation of derived images will occur in this case.



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The interfaces between Viz and the other subsystems are defined at a high level through the following table. This table describes the major interfaces in terms of the requests for information and the corresponding response.

| From | To | Request | Response | Interface Standard |
|---|---|---|---|---|
| Visualization | Host Platform | Insert Images<br>• Secondary Screen Captures<br>• Functional Images<br>• 3D projections (MIP, 3D)<br>• MPR Images<br>• Synthesized contrast-weighted images | Insertion status (from OS)<br>Disk Full<br>Ok | Import into database, (AiM, DMF, or place in import directory.) |
| Visualization | Host Platform | Request for paths of current Selection | Browser Selection information | AiM (Selection File) |
| Visualization | Host Platform | Request next/prior exam/series from current exam/series | Browser next/prior exam/series information (Paths) | AiM |
| Host Platform (Inline) | Visualization | Request (processing type) to generate images using existing images (path) | Resulting functional image/map. | Shared library, cmd line |
| Visualization | Host Platform (DICOM parsing toolkit) | Request for reconstructed images and associated patient info | DICOM Images | DICOM Part 10 Objects |

### 5.4.15. Host & MR Tools [S]

The service tools that interact with the scanner for data collection do so using the SVAT interface via the ATP engine which provides a script based interface. This script based interface provides a means for programmatic based scanner control. Service SW uses this interface to calibrate the Scanner.

The interface provides the following high level control capability via commands:



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- start & end exams
- load protocols
- set scan parameters
- commit scan parameters (i.e. save series)
- download & prep
- modify & retrieve control parameters
- start/stop Prescan
- start/stop scan

The following table summarizes the communications between the subsystems.

| From | To | Data | Interface Standard |
|---|---|---|---|
| MR Tools | Host | SVAT Command + data | SVAT |
| Host | MR Tools | Command Status + requested data | SVAT |

### 5.4.16. MR Tools & Host Platform [T]

MR Service Tools accesses images from the database as needed for calibration, diagnostics, and analysis.

The following table summarizes the communications between the subsystems.

| From | To | Data | Interface Standard |
|---|---|---|---|
| MR Tools | Host Platform | image data request | UAMI |
| Host Platform | MR Tools | image data | UAMI & file I/O |

### 5.4.17. Host & External Scanner Control Device [X]

Host provides a programmatic interface to perform some scanner functions. This interface allows authenticated external devices to operate the scanner using commands and to receive images as they are generated.

An external device is considered authentic only if it is configured on the scanner and is connected to the scanner's internal network. Host authenticates the external connection using the key, hostid of the external device and the IP Address of the external device.

The following operations are supported in this interface:

- Workflow operations like start session, load protocol, save series etc.
- Scanner operations like download, pre-scan, prep, scan etc.
- Parameter modifications like modify CV, set FOV etc.

The Host external interface also supports commands that control the types of images to be installed into the image database. The types of images currently supported are:

- magnitude
- phase
- real
- imaginary

The following table summarizes the communications between the subsystems.

| From | To | Data | Interface Standard |
|---|---|---|---|
| Host | External Device | image data | TCP/IP |
| Host | External Device | responses to requests (status & data) | TCP/IP |
| External Device | Host | commands and status requests | TCP/IP |
| External Device | Host | image data registration | TCP/IP |

### 5.4.18. Host Platform & External Scanner Control Device [Y]

Host Platform supports the following operations for authenticated external devices

- Access to images
- Access to configuration data

The External Device can communicate with Host Platform to obtain images from the scanner database as well as to add images to the scanner database. The images are normally accessed using the standard DICOM communication protocols (C-FIND, C-MOVE, C-STORE) but may also be accessed directly via SFTP (get only)

The External Device may also retrieve configuration data via the SFTP protocol (get only)

### 5.4.19. MR Recon & External Scanner Control Device [Z]

The following interfaces are supported for real-time data and control access between MR Recon and External Devices

- k-space data via the raw data server component
- k-space and control data via the RTP client interface

The raw data server is a standard component of MR Recon. It supports both TCP/IP and InfiniBand connections for streaming raw acquired data to another server, such as the external scanner control device

The RTP client is an application specific component which can use TCP/IP protocols to communicate with processes on other servers, such as the external scanner control device

The following table summarizes the communications between the subsystems

| From | To | Data | Interface Standard |
|---|---|---|---|
| MR Recon | External Device | k-space data | TCP/IP |
| External Device | MR Recon | registration for k-space data | TCP/IP |

## 6. FUNCTIONS

This section characterizes the significant functionalities performed using the System. The Processing Sections cover the normal uses of the System. The additional sections cover common functionalities that apply to all uses, such as managing abnormal conditions (Error Handling & Recovery) and providing UIs in various languages (Internationalization & Localization).

## 6.1. Primary Processing

The following subsections outline the functionalities exercised to fulfill the Scanner's Primary Uses. Each subsection illustrates the relevant subsystems and interfaces, and describes the important processing steps performed for the identified use.

These primary uses are applied for clinical scanning and for some service scanning, such as for diagnosis and calibration. When these uses are applied for service operations, the Patient ID is set to the Service ID. Service operations may be performed manually (directly controlled by service engineers) or through automation (under the control of a service tool initiated by the service engineer).

### 6.1.1. Begin Exam

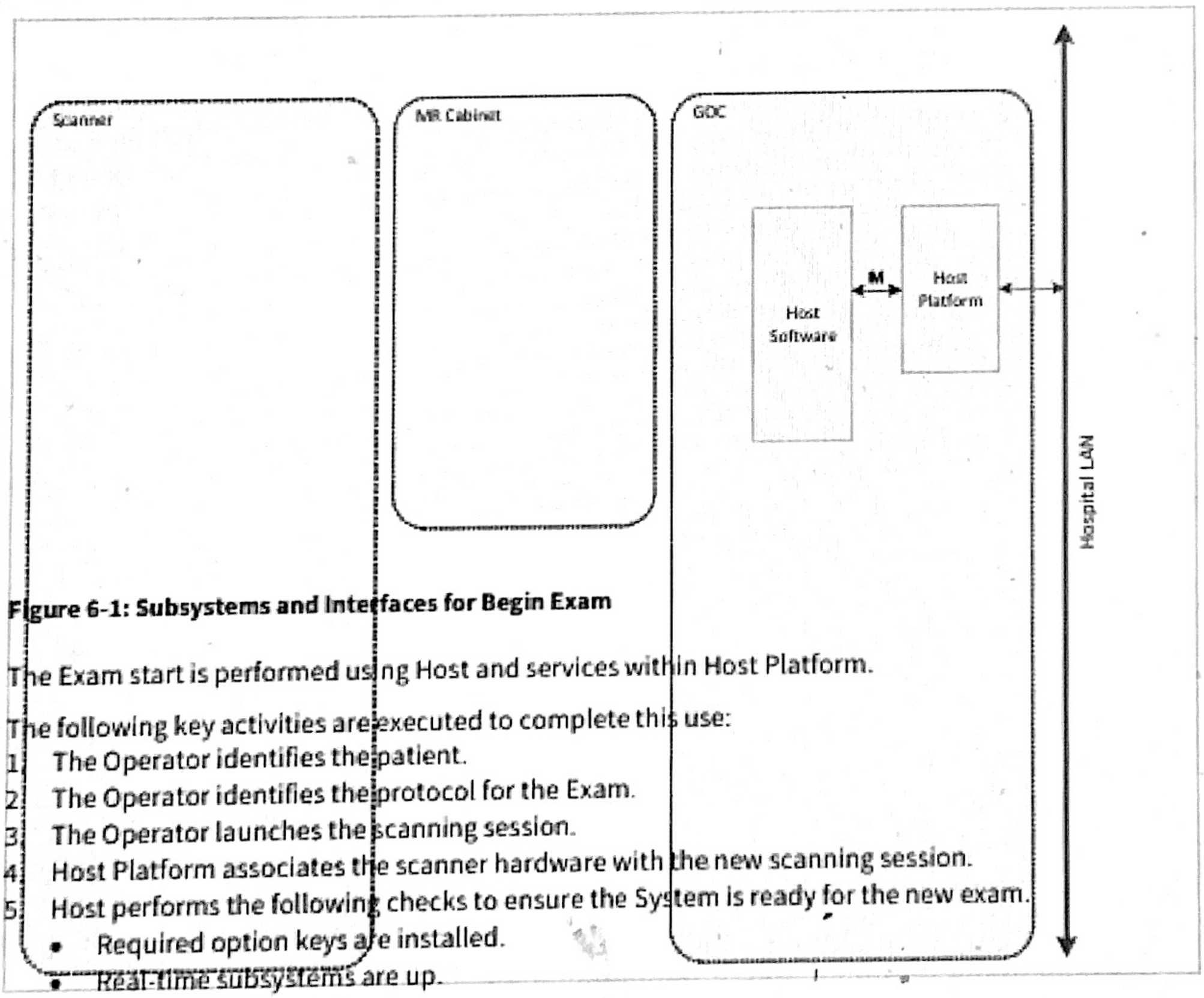



**Figure 6-1: Subsystems and Interfaces for Begin Exam**

The Exam start is performed using Host and services within Host Platform.

The following key activities are executed to complete this use:

1. The Operator identifies the patient.
2. The Operator identifies the protocol for the Exam.
3. The Operator launches the scanning session.
4. Host Platform associates the scanner hardware with the new scanning session.
5. Host performs the following checks to ensure the System is ready for the new exam.
   - Required option keys are installed.
   - Real-time subsystems are up.
   - Scanner is calibrated.
   - No exam/diagnostic session is in progress.
6. After the Exam is started, Host Platform creates a new session UI and displays the appropriate task operations.



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7. Host Platform communicates the patient information to Host.
8. Host sets the safety limits requested by the user.

### 6.1.2. Prescribe Scan

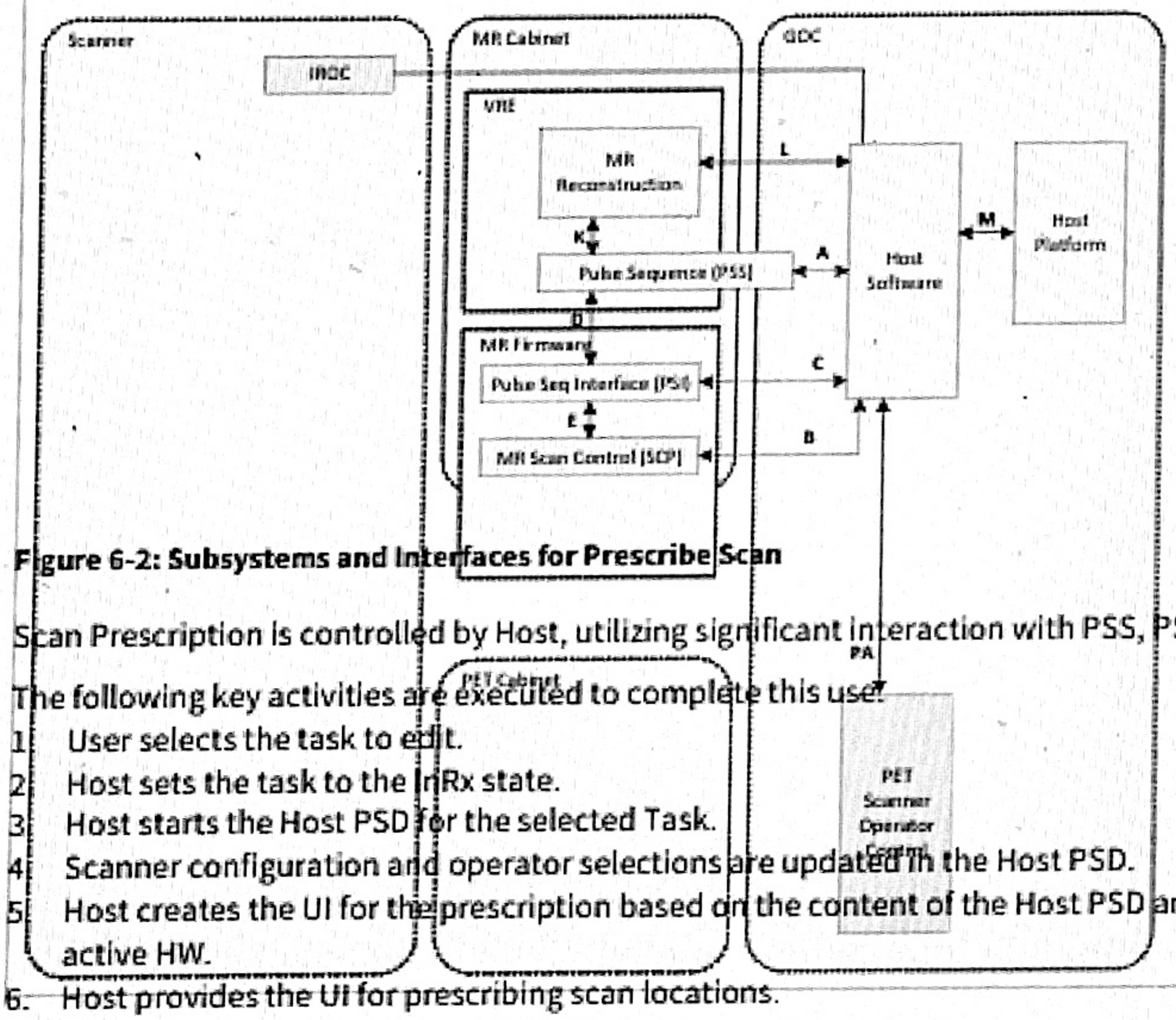



**Figure 6-2: Subsystems and Interfaces for Prescribe Scan**

Scan Prescription is controlled by Host, utilizing significant interaction with PSS, PSI, and SCP.

The following key activities are executed to complete this use:

1. User selects the task to edit.
2. Host sets the task to the InRx state.
3. Host starts the Host PSD for the selected Task.
4. Scanner configuration and operator selections are updated in the Host PSD.
5. Host creates the UI for the prescription based on the content of the Host PSD and the active HW.
6. Host provides the UI for prescribing scan locations.
7. Scan Parameters are selected, including:
   - Coils (RX / TX)
   - Scanning range (slices, slice groups, volumes...)
   - Saturation bands
   - Imaging parameters
   - Post-processing (filtering, subtractions, etc.)

   a. User selects parameters.
   b. Host selects parameters (e.g. Auto-Coil selection).
   c. MR Recon selects parameters (e.g. Auto-Geometry selection).
8. Selections are validated with the Host PSD, and operator is guided with advisory values.
9. Operator saves the prescription.
10. Host sets the task in the RXD state.



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### 6.1.3. Initiate Scan

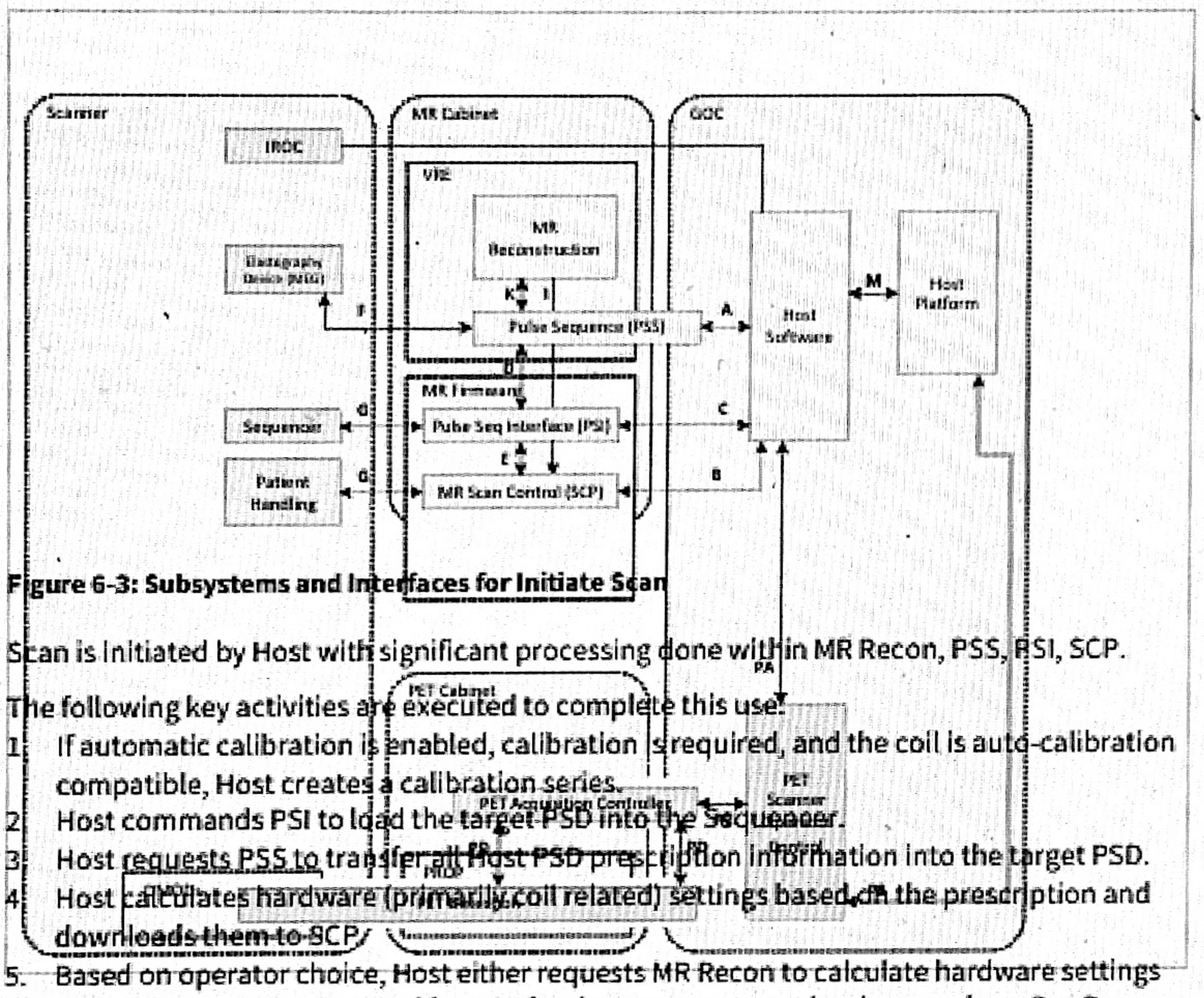



**Figure 6-3: Subsystems and Interfaces for Initiate Scan**

Scan is initiated by Host with significant processing done within MR Recon, PSS, PSI, SCP.

The following key activities are executed to complete this use:

1. If automatic calibration is enabled, calibration is required, and the coil is auto-calibration compatible, Host creates a calibration series.
2. Host commands PSI to load the target PSD into the Sequencer.
3. Host requests PSS to transfer all Host PSD prescription information into the target PSD.
4. Host calculates hardware (primarily coil related) settings based on the prescription and downloads them to SCP.
5. Based on operator choice, Host either requests MR Recon to calculate hardware settings for maximum output or provides a UI for the operator to set hardware values. See Prescan function (section 6.1.4) for further details.
6. Host generates image headers based on user prescription values and communicates image reconstruction parameters to MR Recon.
7. Based on operator choice, start/stop/pause of scanner can be initiated from user interface or hardware control. In both cases, scanner state is communicated between Host and SCP via MR Recon.



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### 6.1.4. Prescan

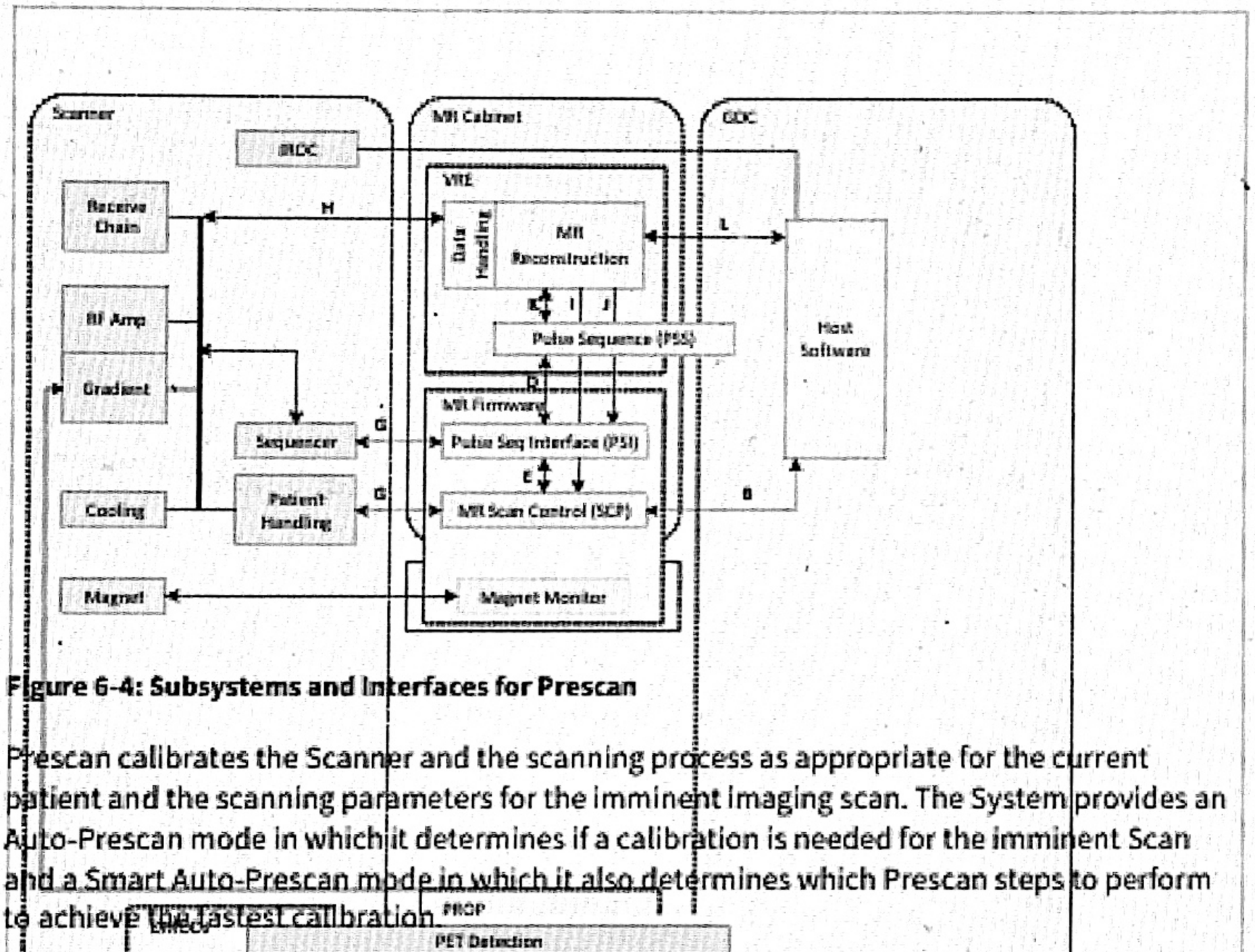



**Figure 6-4: Subsystems and Interfaces for Prescan**

Prescan calibrates the Scanner and the scanning process as appropriate for the current patient and the scanning parameters for the imminent imaging scan. The System provides an Auto-Prescan mode in which it determines if a calibration is needed for the imminent Scan and a Smart Auto-Prescan mode in which it also determines which Prescan steps to perform to achieve the fastest calibration.

Prescan is initiated by Host, which triggers MR acquisition through SCP, which in turn triggers PSI to drive the Scanner, which produces data collected by MR Recon.

Prescan receives data frames from MR Recon and applies algorithms on the received data to compute:

- Transmit Gain (TG)
- Center Frequency (CF)
- Analog Gain (R1)
- Digital Gain (R2)
- autoshim parameters
- Receiver Noise weights
- Coil Calibration
- ASSET/PURE Calibration

When operating in Manual mode, Prescan formats the calculated values, provides a UI to display them on the GOC, and allows the operator to alter or confirm the values. When operating in Auto mode, Prescan applies the calculated value without user intervention.

For all MR acquisition on the Scanner, the Magnet Monitor is engaged and disables MR Scanning if it detects any danger to the Magnet.

Refer to the Prescan SSDD [41] for further details on the Prescan architecture and design.


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### 6.1.5. Execute Scan (Acquire & Reconstruct)

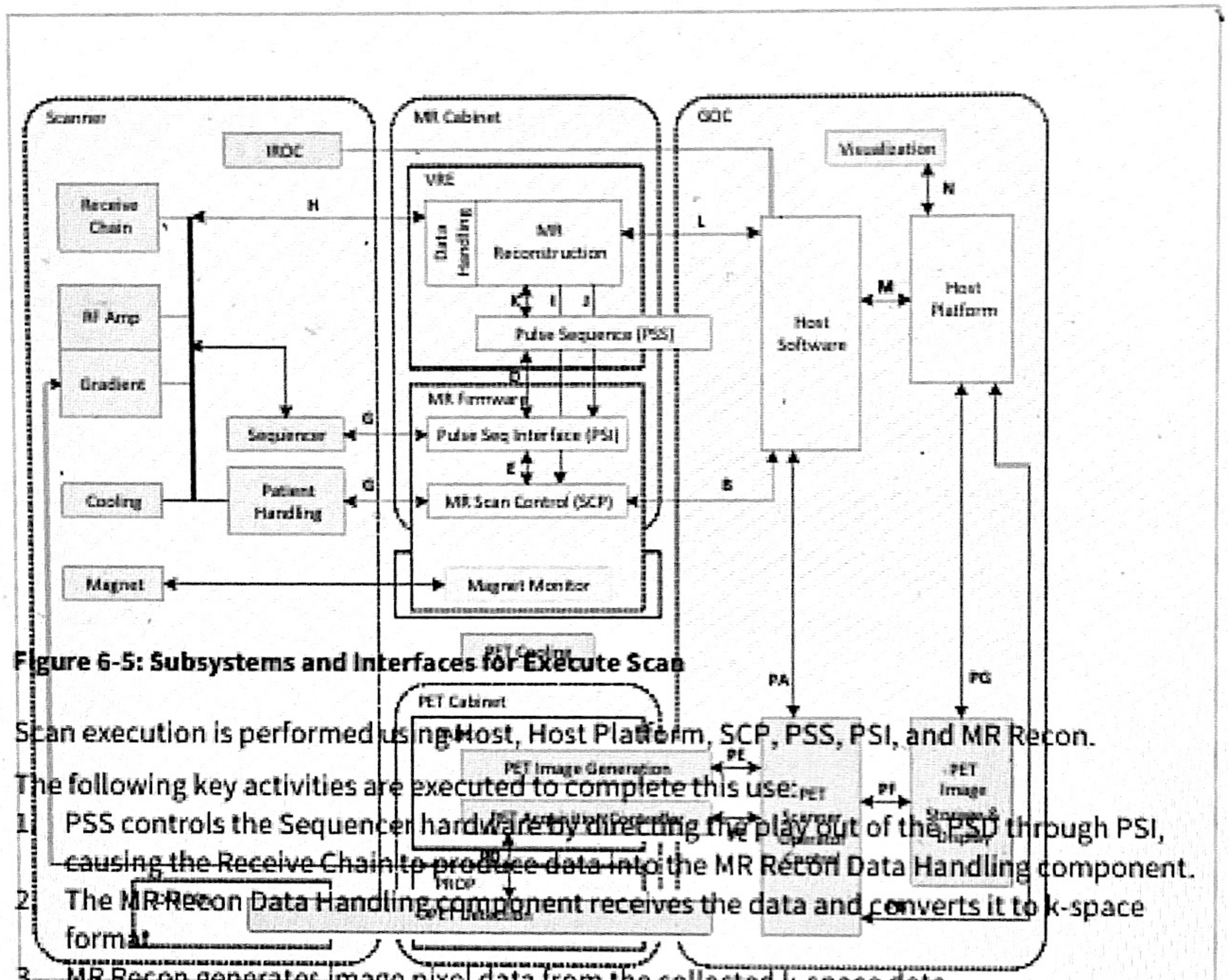



**Figure 6-5: Subsystems and Interfaces for Execute Scan**

Scan execution is performed using Host, Host Platform, SCP, PSS, PSI, and MR Recon.

The following key activities are executed to complete this use:

1. PSS controls the Sequencer hardware by directing the play out of the PSD through PSI, causing the Receive Chain to produce data into the MR Recon Data Handling component.
2. The MR Recon Data Handling component receives the data and converts it to k-space format.
3. MR Recon generates image pixel data from the collected k-space data.
4. MR Recon transfers the image pixel data to Host.
5. Host receives the pixel data and generates DICOM images by combining the applicable image header parameters with the pixel data.
6. Host Platform stores the DICOM images in the image database.
7. For real-time applications, Host displays the acquisition management UI allowing the operator to monitor and adjust the scan.

For all MR acquisition on the Scanner, the Magnet Monitor is engaged and disables MR Scanning if it detects any danger to the Magnet.



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### 6.1.6. Inline Process and View

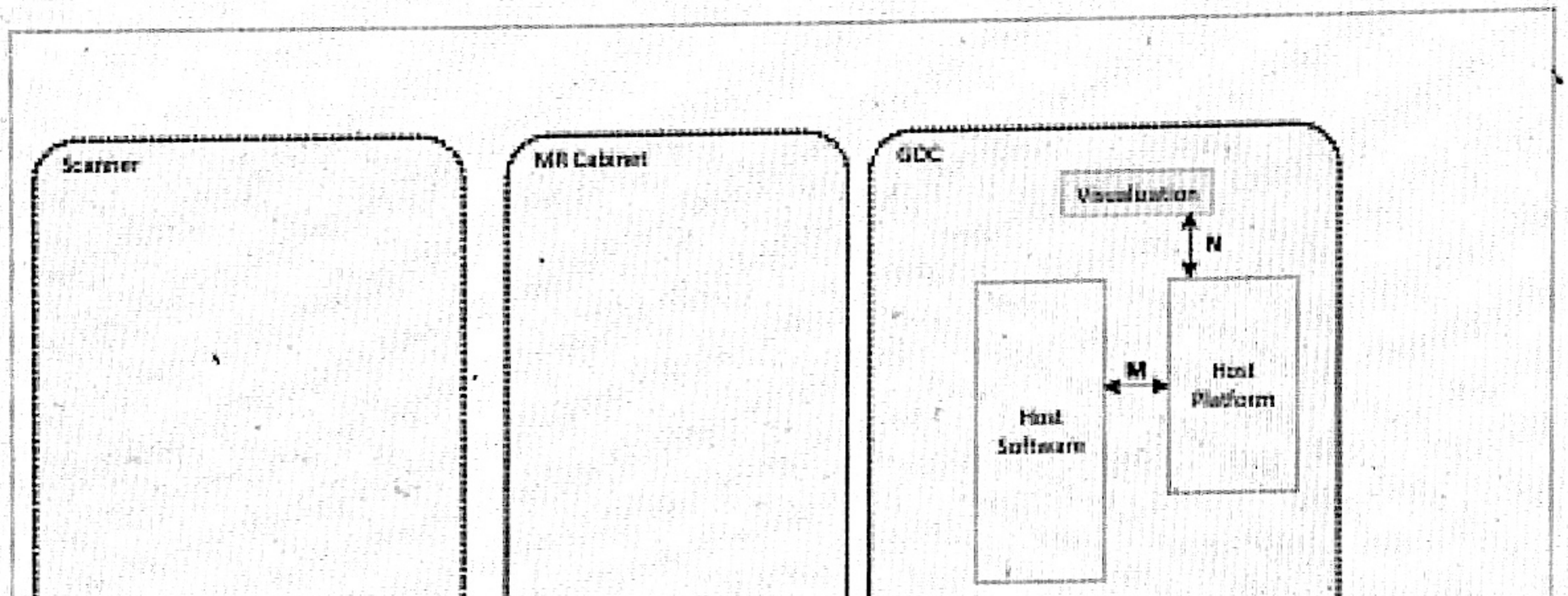



**Figure 6-6: Subsystems and Interfaces for Inline Process and View**

The Inline Post Processing passes the post processing and viewing initiated by Host through services provided by Host Platform and Visualization.

The following key activities are executed to complete this use:

1. The automatic post-processing subtask queues up a job to the visualization service.
2. The visualization service registers the job with the visualization session manager to initiate GEOCOMM communication.
3. The visualization session manager returns the information for the GEOCOMM connection formed for communication between the subtask and the visualization service carrying out the subtask.
4. Communication on statuses, occur between visualization service and the subtask.
5. For some specialized applications, data is sent back to subsequent acquisition tasks (e.g. Ready Brain – spatial Rx, B1 Optimizer – phase/amplitude.)
6. The visualization service terminates.

The manual inline processing launches a visualization application like MR Pasting or Clariview and simply monitors the launched process to determine when the application has terminated. A thread is allotted to the application to monitor the application process termination.



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### 6.1.7. End Exam

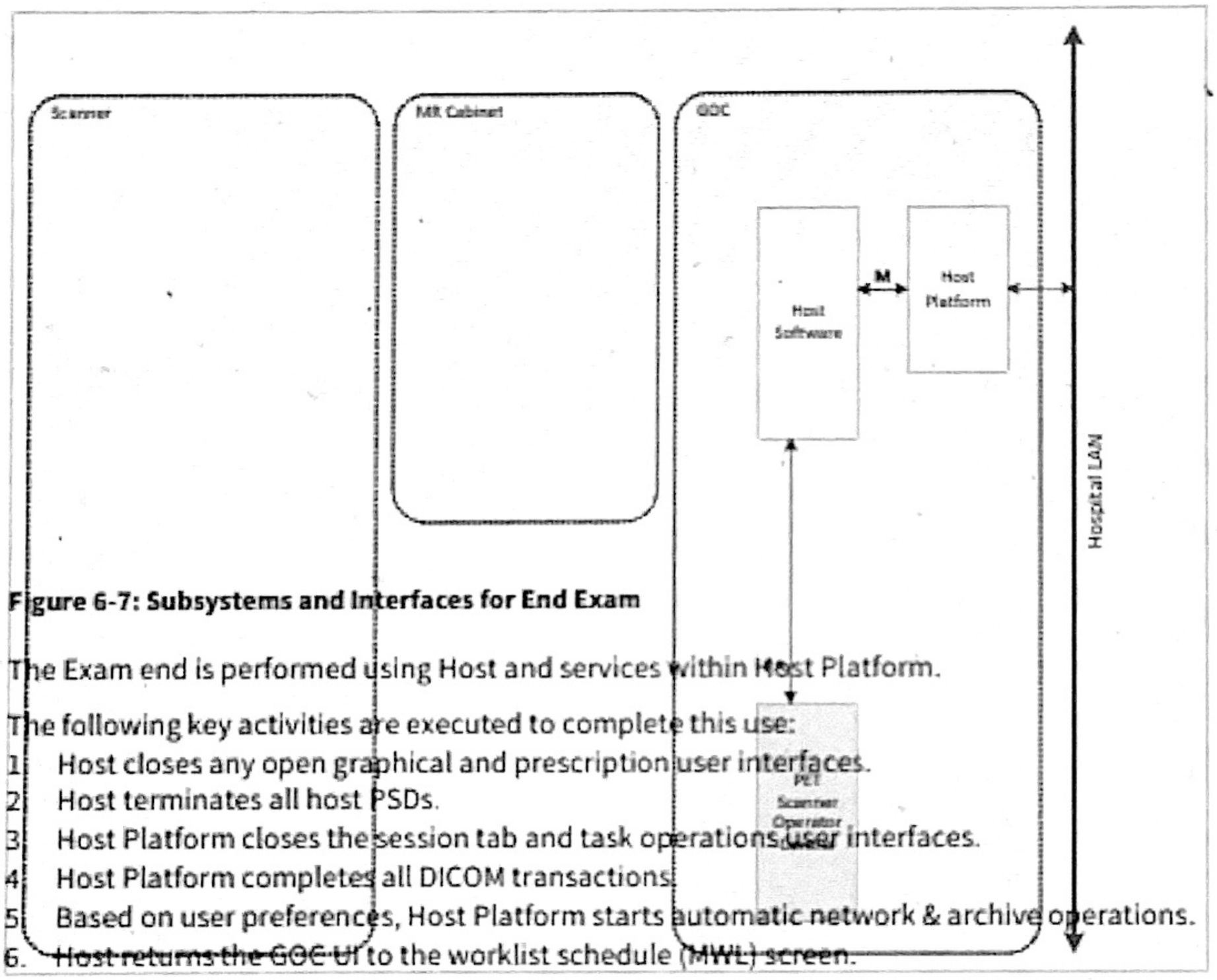



**Figure 6-7: Subsystems and Interfaces for End Exam**

The Exam end is performed using Host and services within Host Platform.

The following key activities are executed to complete this use:

1. Host closes any open graphical and prescription user interfaces.
2. Host terminates all host PSDs.
3. Host Platform closes the session tab and task operations user interfaces.
4. Host Platform completes all DICOM transactions.
5. Based on user preferences, Host Platform starts automatic network & archive operations.
6. Host returns the GOC UI to the worklist schedule (MWL) screen.



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### 6.1.8. Post-Process and View

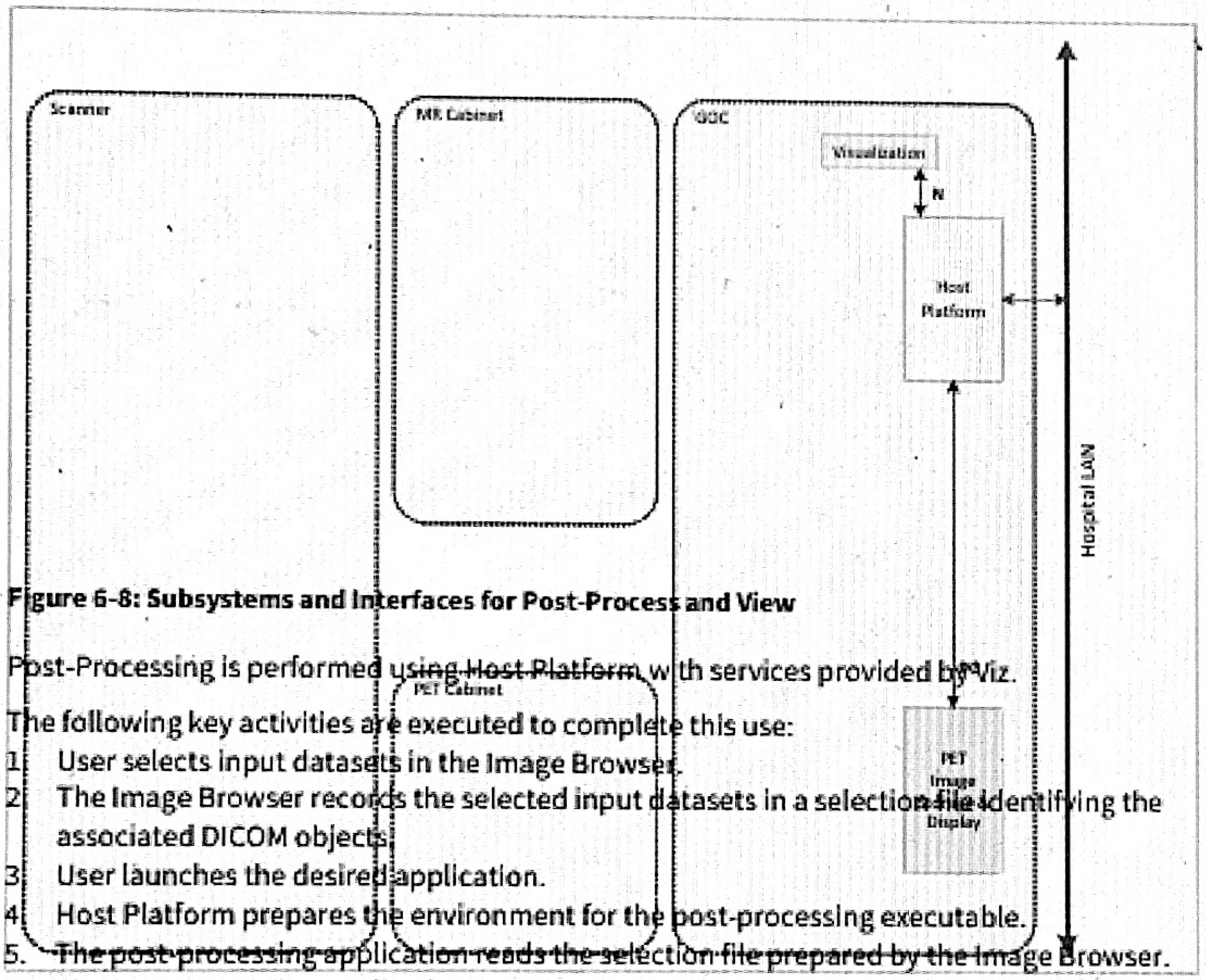



**Figure 6-8: Subsystems and Interfaces for Post-Process and View**

Post-Processing is performed using Host Platform with services provided by Viz.

The following key activities are executed to complete this use:

1. User selects input datasets in the Image Browser.
2. The Image Browser records the selected input datasets in a selection file identifying the associated DICOM objects.
3. User launches the desired application.
4. Host Platform prepares the environment for the post-processing executable.
5. The post-processing application reads the selection file prepared by the Image Browser.
6. The post-processing application validates the selections to determine if the appropriate and necessary datasets are selected.
7. The post-processing application performs its workflow, including loading into internal data structures the images associated with the selected input datasets.

Post-processing applications may display UIs and accept user inputs to guide the analysis. User inputs are typically applied to 1) direct the type of analysis to perform, 2) set analysis parameters, or 3) identify the form in which results are generated (e.g. screen-save objects, processed images, 3D projection images).



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Figure 6-9 illustrates the data-flow for a typical post-processing session.

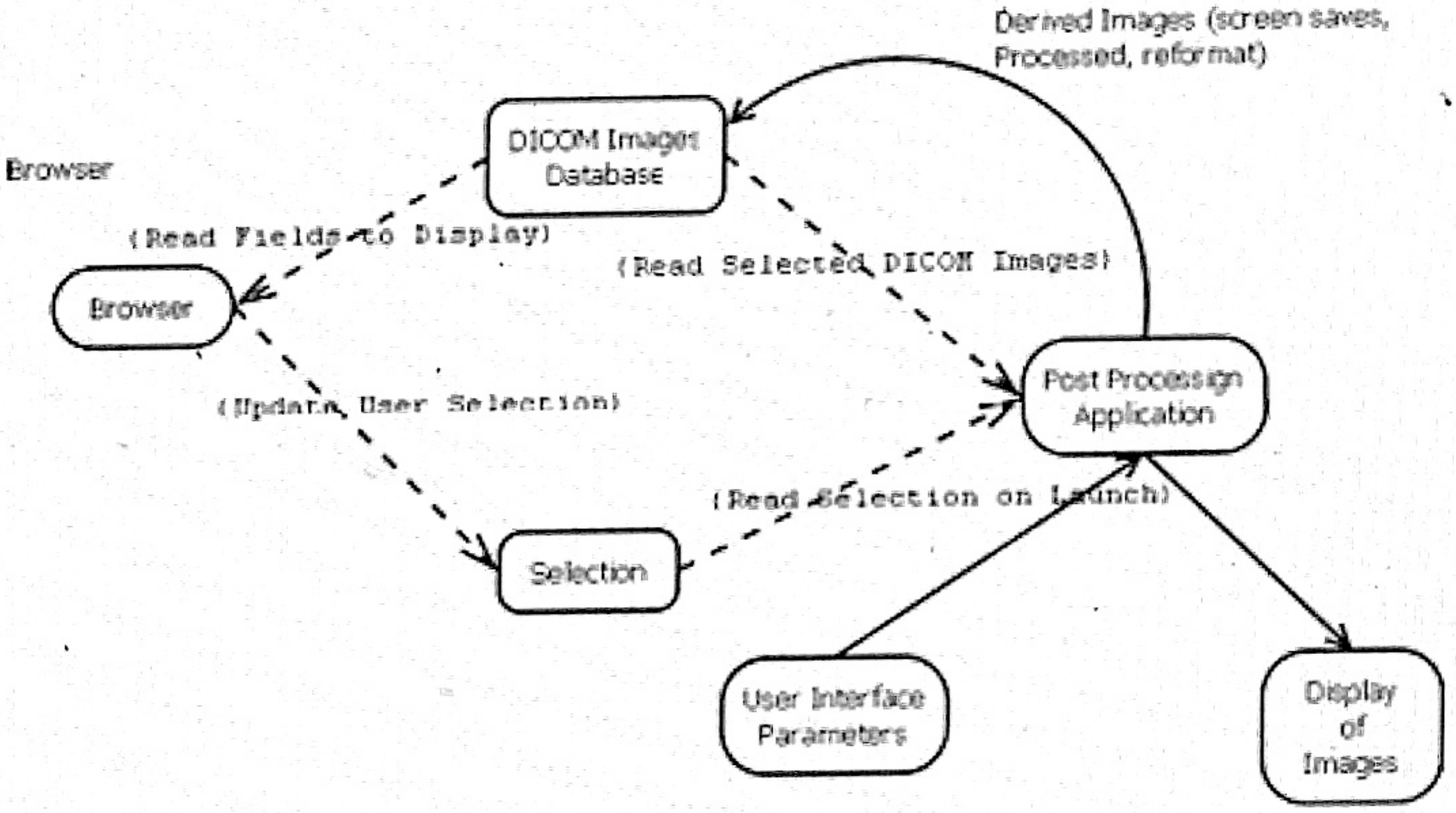



**Figure 6-9: Post-Process Data Flow**

## 6.2. Secondary Processing

The following subsections outline the functionalities exercised to fulfill the Scanner's Secondary Uses. Each subsection illustrates the relevant subsystems and interfaces, and describes the important processing steps performed for the identified use.

### 6.2.1. Install System

MR Tools is responsible for installation. Portions of the Software run stand-alone in order to install the operating system and custom software.



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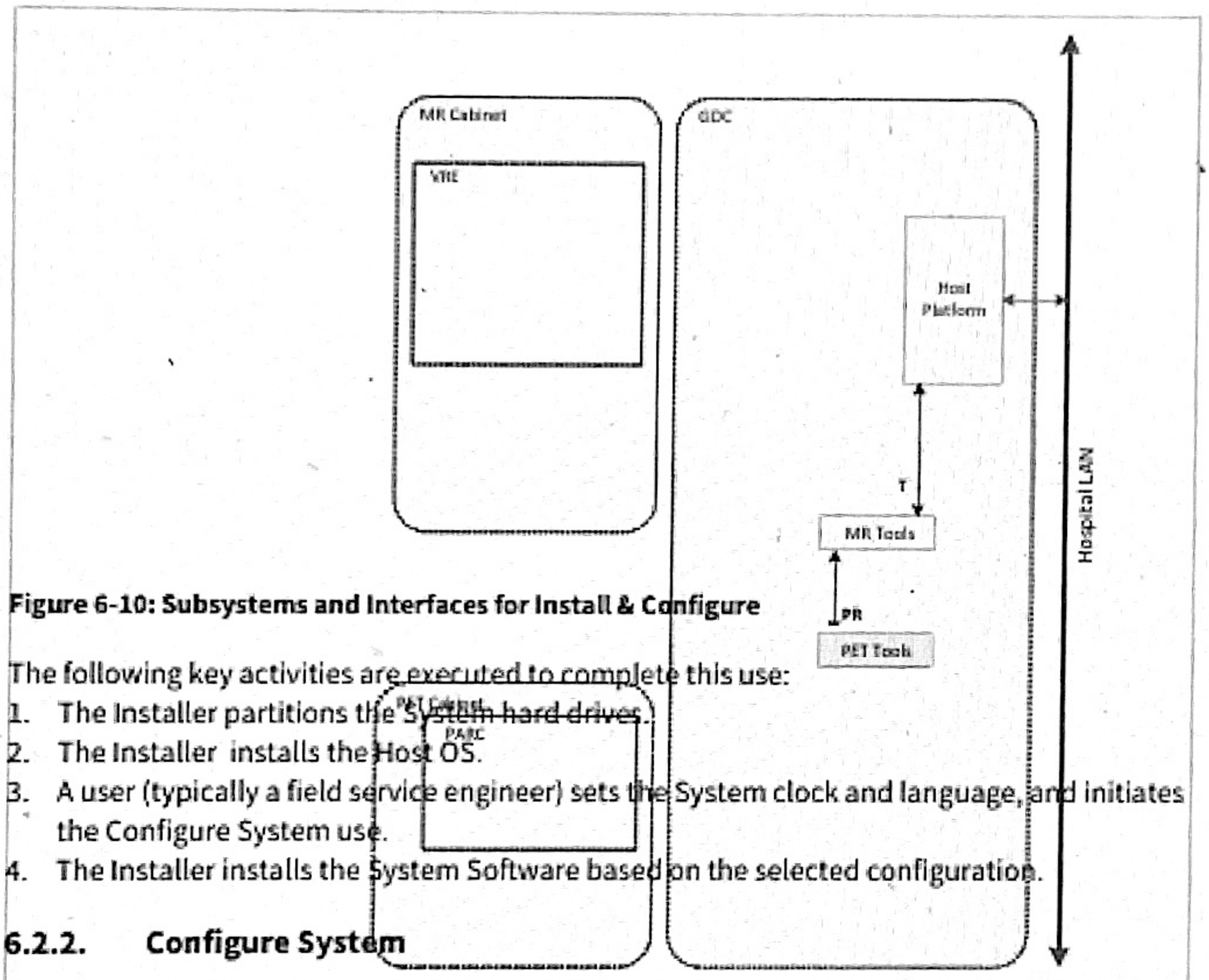



**Figure 6-10: Subsystems and Interfaces for Install & Configure**

The following key activities are executed to complete this use:

1. The Installer partitions the System hard drives.
2. The Installer installs the Host OS.
3. A user (typically a field service engineer) sets the System clock and language, and initiates the Configure System use.
4. The Installer installs the System Software based on the selected configuration.

### 6.2.2. Configure System

MR Tools is responsible for configuration. Portions of the Software run stand-alone to set up the Scanner for the specific hardware and features associated with a particular variant of the Scanner.

The following key activities are executed to complete this use:

1. The user configures the network, hospital name, and other site preferences, and specifies the hardware configuration.
2. The user installs licenses to enable optional features allocated to the site.



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### 6.2.3. Calibrate System

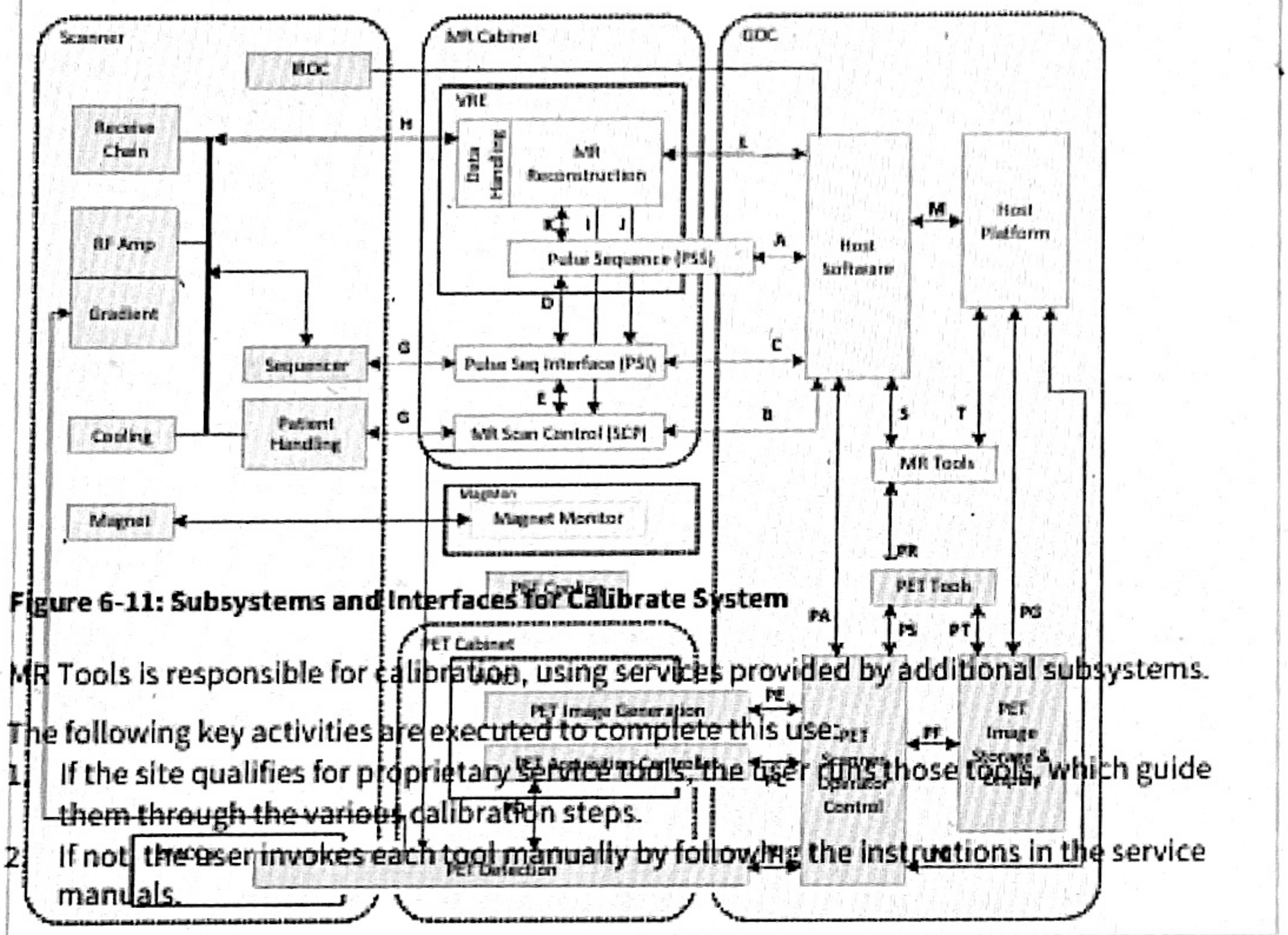



**Figure 6-11: Subsystems and Interfaces for Calibrate System**

MR Tools is responsible for calibration, using services provided by additional subsystems.

The following key activities are executed to complete this use:

1. If the site qualifies for proprietary service tools, the user runs those tools, which guide them through the various calibration steps.
2. If not, the user invokes each tool manually by following the instructions in the service manuals.

### 6.2.4. Start Up System

System Initialization includes the software operations performed from Scanner power-on to the point where the operator may begin an acquisition or post processing session on the Scanner. This initialization phase includes initial operator login for systems not configured for auto-login as well as the operator login to unlock the screen and gain access to the Software.

The first phase of startup is performed by each machine's boot process and is managed via scripts. This phase launches the System Layer, starting with the OS services (such as networking) as appropriate for the Scanner's configuration. In addition, the coil database server is started and loaded. These scripts also start the Host Monitor.

The second phase of startup is managed by the Host Monitor. This phase launches each Service Layer process based on its configuration file. Each configuration specifies the startup conditions (dependencies, environment, parameters) as well as monitoring options (watchdog period, shutdown, restart) for a process. During this phase, NSP is started. NSP cleans up any temporary files remaining from the previous run and verifies sufficient disk space exists for proper Scanner operation. It checks the integrity of the Scanner configuration files and any corrupted files are replaced by their backup versions. NSP manages the real-time system (MR Recon and SCP) startup. It resets the SSC chassis and the ICNs, and then monitors


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the bootup of those machines, including the waits for the machines to obtain network addresses (via DHCP). NSP starts the Nuevo platform services and the window manager. NSP notifies the Host Monitor and the remaining processes (those with NSP as a precursor) are started. At the end of the second phase, the Scanner is ready for user interaction.

The third phase of startup is initiated by user login, either explicitly by a user or via the auto-login mechanism. This phase launches parts of the Application Layer.

### 6.2.5. Shut Down System

System termination includes the software operations performed to complete an orderly shutdown of the Scanner. System shutdown can be caused either by the failure of a critical system component, detected by host monitor, or manually initiated by the operator.

In all cases, the shutdown progresses by closing all open sessions, stopping processes dependent on managed processes from host monitor, and signaling host monitor to stop. Host monitor resets SSC, and stops all its managed processes (in the order specified via stop precursors in the initialization files), including stopping NSP, which performs an orderly power off of the VRE and stops all remaining Nuevo services.

For systems with Cabinet Monitor hardware, the Cabinet Monitor software is notified to expect HOST to turn off and to extend system power by any additional required cooling time. Once all user-level processes have stopped, the latent IPC and temporary files are removed, the X-window manager is stopped, and the HOST shutdown is initiated. For systems with Cabinet Monitor hardware, the Cabinet Monitor software detects HOST power is turned off, waits the additional cooling time it was previously advised, and then turns off system power.

### 6.2.6. Monitor Magnet

The Magnet Monitor works independently of the rest of the Scanner. It does not communicate with other computerized portions of the Scanner.

## 6.3. Error Handling and Recovery

The Software maintains a System Log which provides operators, field engineers, and development engineers with an operational history of the Scanner, including information regarding warning and error conditions and a variety of events. When records are registered to the System Log, they are encrypted for storage; access to System Log records is controlled using service licenses (aka SSA Licenses). The Software provides a System Log Viewer to facilitate user access to the information. Additional application-specific log files record detailed information that would be inappropriate for the System Log.

The Software detects failures of many system processes and if it detects a process has failed, the Software either performs an automatic restart of the failed process or notifies the operator of the failure and optionally invokes the Shut Down System function. The action taken depends on the criticality of the failed process as well as the ability of other cooperating processes to handle the failure and a subsequent restart of the process.



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The Software allows users to perform a partial shutdown and restart (reset) of the Scanner in order to recover from failures, minimizing the time impact for the recovery and sometimes allowing the user to continue an exam that is in progress.

The reset process terminates and uses the startup infrastructure to restart all subsystems which are external to HOST. Each HOST-based subsystem is responsible to handle the reset event. Details of the reset (startup) actions are covered in the detailed design for each subsystem.

## 6.4. Internationalization and Localization

The Software supports the display and entry of text in multiple natural languages as described in the Regulatory FRS [20]. There are several aspects involved with developing Software that contains text in multiple languages (Internationalization, aka I18n) and applies the text of one of the available languages for a Session (Localization), including:

- Designs to separate text strings from business logic
- Source code Organization to isolate text strings of each supported language
- Procedures to Make/Build/Package text in all supported languages
- Methods to Configure Session to apply a single supported language
- Mechanisms to Render text of the selected language

The Software Architecture considers the aspects above for all functionalities intended to be used by Technologists and Reading Physicians, but not for functionalities intended to be used exclusively by Field Service Engineers or Manufacturing Technicians. The aspects above are fulfilled differently depending on the functionality and the primary programming language used to implement the functionality. The functionalities that apply Internationalization/Localization include:

- UIs for Scanning (Prescribe, Acquire, Reconstruct, Monitor)
- UIs for Viewing and Post-Processing Images
- UIs for Managing Exams, Patients, Protocols, Tracers, Images
- UIs for Qualifying the System (Daily QA)
- UIs for Resetting Hardware
- UIs for Examining the System Log
- UIs for Authentication (Log In, Log Out) and Managing Security
- Reports for any of the above
- Online Help for any of the above

The primary programming languages used to implement the above functionalities include:

- Java (with/without Swing)
- C/C++ (with/without Motif)
- Objective C
- HTML



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# 7. DATA

The data managed by the Software falls into the following general categories: static data storage, disk utilization for dynamic data, acquisition memory utilization and host memory utilization.

## 7.1. Static Data

The primary static data and storage managed on the Scanner fall into the following general categories: configuration data, calibration data, and protocol data.

## 7.2. Dynamic Data

The primary dynamic data managed on the Scanner is comprised of raw data (k-space data) and image data (DICOM images). During acquisition, multiple transient datasets are generated progressing toward the final DICOM images. Additionally, a small amount of dynamic data is associated with the patient schedule (local storage for MWL queries and local schedule items.)

## 7.3. Acquisition Memory

Acquisition memory is used to hold raw data prior to reconstruction, intermediate reconstructed image data and image data prior to transport to the host for DICOM image generation and storage.

## 7.4. Session Memory

Session memory is utilized by the set of cooperating processes managing the scanner operation as well as the transient processes associated with individual acquisition tasks and post processing tasks. In order to control the resource utilization, the number of post processing sessions has been bounded and inline post processing tasks work in a request/use/release memory mode.

## 7.5. Data Management

The disk subsystem is partitioned to provide separate storage areas for image data and raw data.

When preparing for a scan the Scanner verifies that sufficient acquisition memory is available and reserves image database space for the prescribed number of images for the scan. If sufficient resources are not available the scan acquisition initialization fails and the user is notified of the reason.

Log files are capped at maximum sizes (per log file defined in the detailed design for each component) and the files are aged to maintain a sliding window of prior log entries for post failure analysis.


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The data items passed between subsystems and their management policies are described further in the detailed designs referenced in section5.4.

# 8. SAFETY SEGREGATION

Figure 8-1 identifies the Safety Class C components that are segregated within the Software.

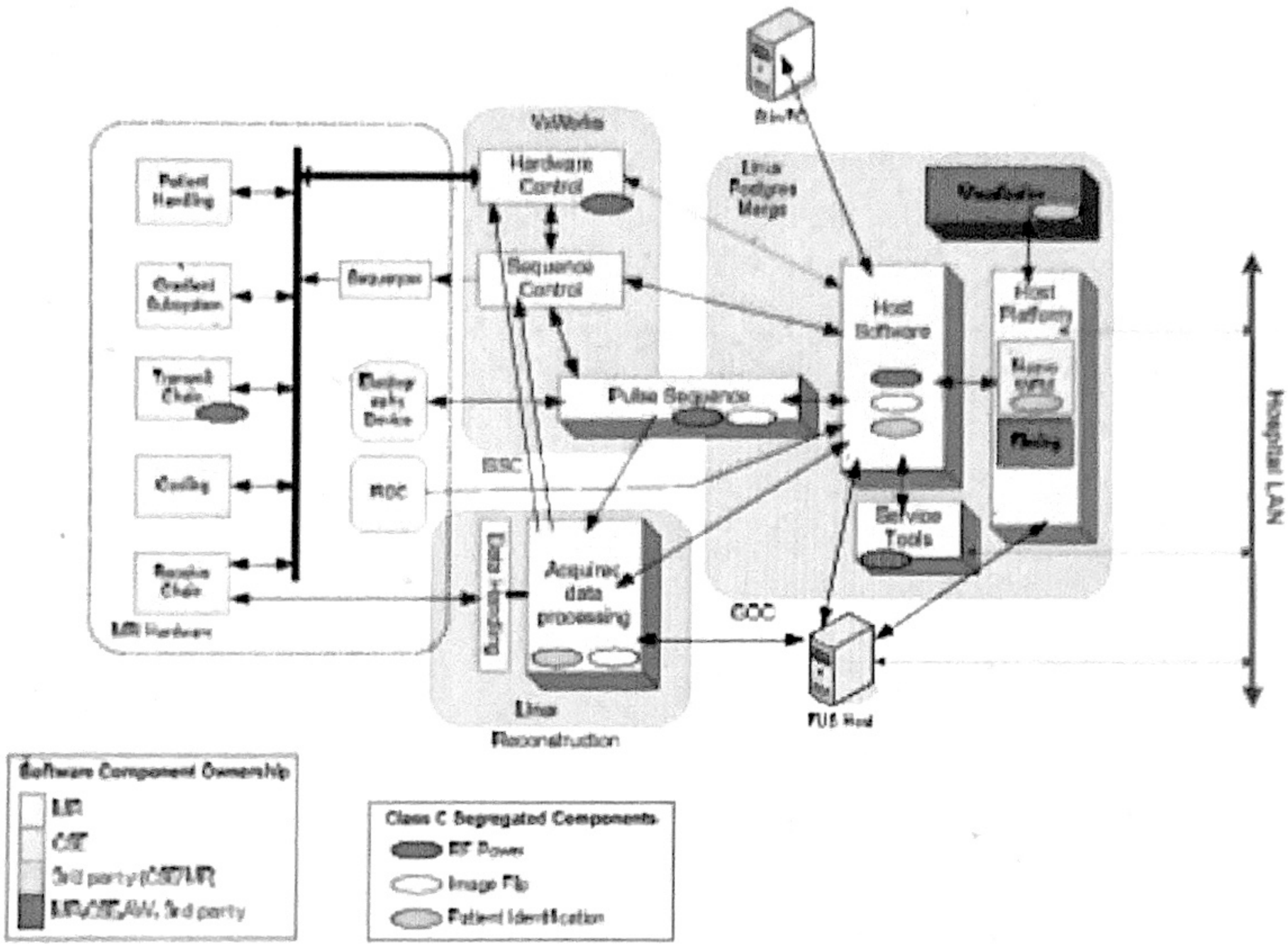



**Figure 8-1: Safety Class C Segregation**

The software associated with the critical hazards has been segregated from the remainder of the software, resulting in a Class B designation of the other components. Further segregation has been performed to separate Class-A components from the Class-B components associated with the minor and severe safety hazards. The detailed component classification is maintained in the SW Structure and Safety Classification [15]. Some components may be indirectly associated with hazards by residing within a process with other safety-related components. In these cases, the component has not been associated with a specific hazard(s). For memory corruption issues, the mitigations are handled on a subsystem level as specified in the subsystem FMEAs.

Critical data (data where misapplication could lead to a Hazard) is encapsulated with a checksum so that any component that applies the data can detect data corruption and take the appropriate action to ensure system, operator and patient safety. This prevents having to assess higher Safety Classifications on components that only handle, but do not apply, critical


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data. Details of the Class-C segregation, for the critical hazards, are contained in the sections below.

The primary segregation between Class-A , Class-B, and Class-C is through isolation. This isolation is managed through the software design ensuring the processes exist as separate executing process w/o shared resources to components of a higher classification. In some cases the segregation has been accomplished by checksum encapsulation, or other means, within the process to guard against/detect memory corruption errors.

## 8.1. RF Power Protection

The RF power management software has been segregated from the surrounding software by means of separation, encapsulation and acknowledgement. The critical user input associated with SAR control is patient weight along with the selected SAR control mode.

The pulse sequence portion of the power monitor RF calibration tool set is the component associated with the pulse sequence which is segregated from the remainder of PSS.

The power monitor is a separate component within the Scanner responsible to monitor the output RF power and to disable the RF transmitter if the transmitted power exceeds the safety settings. The receipt of data parameters transmitted to the power monitor is acknowledged to ensure the proper parameter set is being enforced. The critical RF power parameters are encapsulated via checksums to prevent inadvertent alteration. This data is checked at the endpoints using the information. The SAR Safety component, which was added to compute the setup parameters for the power monitor, runs as an independent component to prevent interaction with the other operations performed during scan prescription. This component validates its input data to ensure that the input has not been altered. Additionally, it verifies the consistency keys within the coil data to ensure the data belongs to the correct coil. Figure 8-2 depicts the dataflow within the Scanner associated with this activity.



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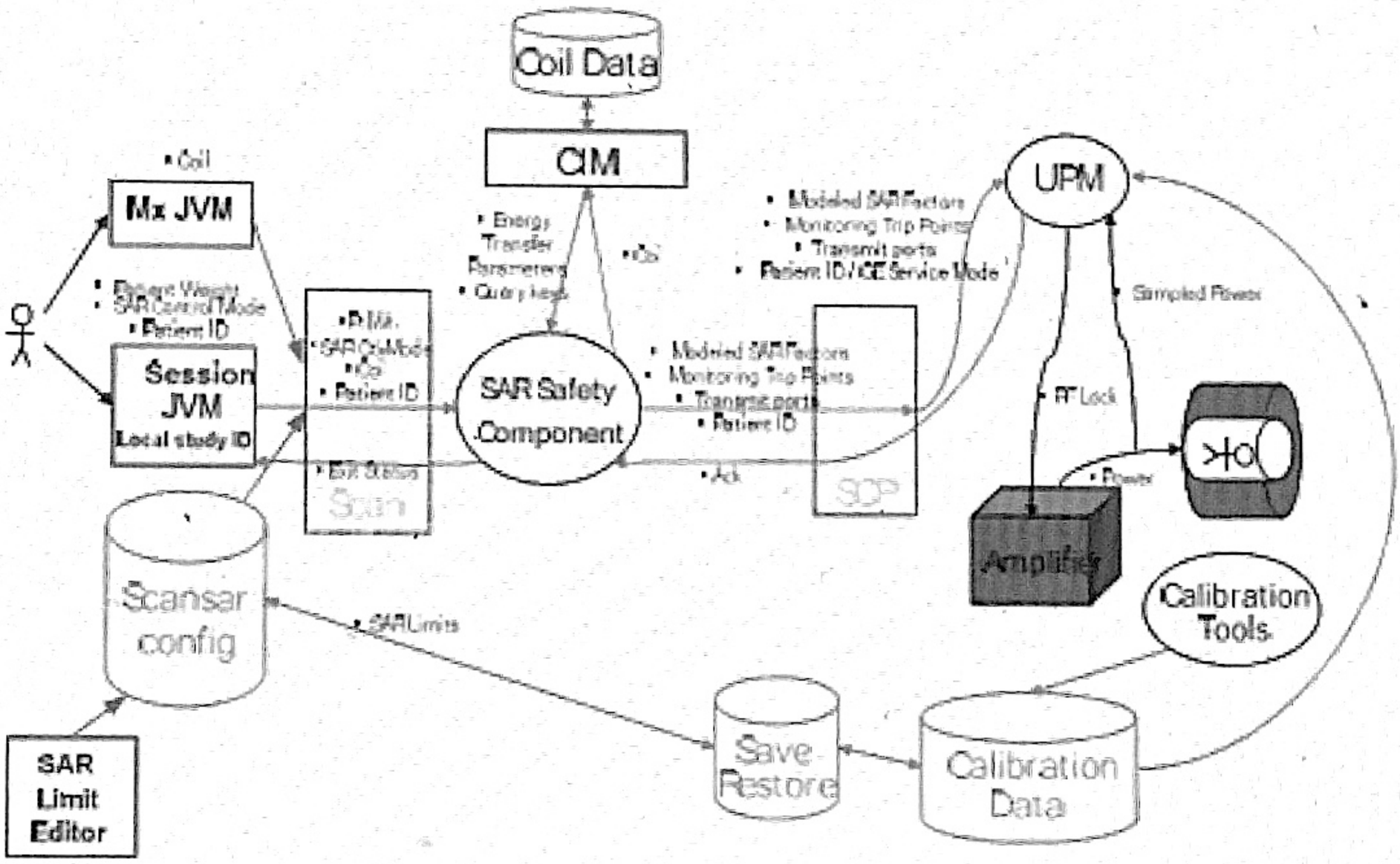



**Figure 8-2: RF Power Setting Data Flow**

The software tools associated with RF calibration are also considered Class-C. When the tool completes the Scanner calibration, the calibration data will be signed with a checksum to ensure that the data integrity is maintained.

## 8.2. Image Flip Protection

The hazards associated with image flips or improper image ordering (that is, incorrect display or annotation of image right/left) are managed via isolation, consistency checks, and checksum encapsulation of the data when it is being transferred between the various system components. In order to protect from unintended image flips the critical components that could cause image flip within each subsystem are isolated. Checksums are used to protect the data integrity as it passes thought the various components within the subsystems.

The critical computations have been isolated into separate libraries (PSD and Host) to segregate them from the remainder of the code set. Data parameters which reside in System configuration files are checked for sign and range. Since the image header data follows a different path through the Software than the pixel data, the process which combines the header and pixel data into the final image also verifies that the pixel data matches the expected spatial data for the image.

Figure 8-3 below shows the critical data flow within the Scanner related to Image flips.


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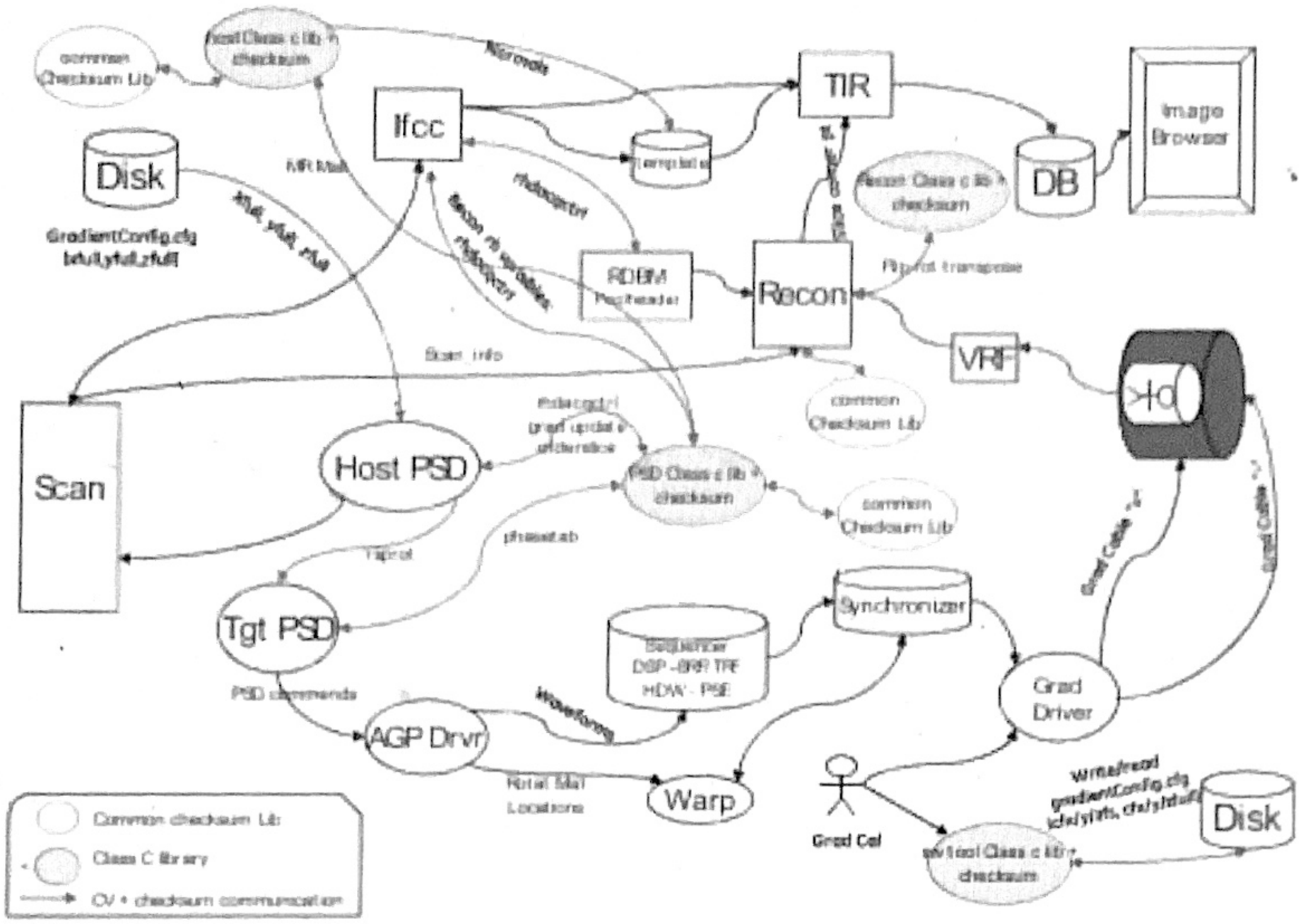



**Figure 8-3: Image Orientation Data Flow**

## 8.3. Patient Identification Protection

The hazard related to patient identification occurs if the incorrect patient information (patient ID, name, date of birth) is assigned to image data. The software components that could contribute to this hazard have been segregated from the remainder of the software.

The data passing between various subsystems is protected through the use of checksums. The subsystem which creates the patient information generates a checksum of the data. This checksum accompanies the data and is rechecked at image installation time to ensure data integrity. Additional integrity checks verifying the exam, series and image numbers are performed at the point where the pixel data is associated with the header data.

MR Recon clears all working memory between studies to ensure image data does not inadvertently cross study boundaries. Additionally, all accesses to working memory within MR Recon are validated to ensure the correct memory segments are being accessed.

The database subsystem ensures that image checksums exist in the DICOM image preamble. Images received via media or network transfers will have checksums added prior to database insertion. Images being installed from within the Scanner are checked against their respective checksum data.

The Viz and Applications software use the embedded checksums to verify that the images have not been corrupted. Additionally, they verify that the set of data received from a



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